

DEPARTMENT OF MATHEMATICS & STATISTICS

MTH 211: Theory of Statistics

(Session: 2025-26 Semester: II)

Problem Set 7 (Testing of Hypotheses – Likelihood Ratio Test)

1. Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$ distribution, where μ and σ are unknown. Suppose that we observe the values: 2.82, 2.71, 3.22, 2.67 and would like to test $H_0: \mu = 2$ against $H_A: \mu \neq 2$.
 - (a) Assuming $\alpha = 0.1$ construct the small sample exact LR test. Does your decision change if the asymptotic test is considered?
 - (b) Now assuming $\sigma^2 = 1$, if we require a significance level α , find β (probability of Type II error) as a function of μ and α in the exact test.
2. Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$ distribution, σ known. Construct the small sample exact LR test of size- α for $H_0: \mu = \mu_0$ against $H_A: \mu < \mu_0$.
3. Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$ distribution, μ, σ unknown. Construct the small sample exact LR test of size- α for $H_0: \sigma^2 = \sigma_0^2$ against $H_A: \sigma^2 \neq \sigma_0^2$.
4. Let $X_1, \dots, X_n$ be a random sample from a continuous distribution with pdf $f_\theta(x)$ where $\theta \in \Theta = \{\theta_0, \theta_1\}$. Consider testing $H_0: \theta = \theta_0$ against $H_A: \theta = \theta_1$. Show that LR size α test leads to the MP size α test.
5. Consider testing $H_0: \theta \geq \theta_0$ against $H_A: \theta < \theta_0$ with a random sample $X_1, \dots, X_n$ from
$$f_\theta(x) = \begin{cases} \theta x^{\theta-1}, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find the small sample size- α LR test based on the exact distribution of the LR statistic.

6. An automobile company wants to check if its best-selling product is equally popular in Delhi, Mumbai and Bangalore. They conduct a survey among a random sample of automobile users in each of the three cities and ask if that particular product is used or not. They collect the following data:

City	New Delhi	Bangalore	Mumbai
Sample Size	100	100	200
No. using the product	14	27	47

Use the LR test at level $\alpha = 0.05$ to test the null hypothesis of an equal popularity of the product in each city against the alternative that it is not.